

Small-world networks and upgrading in Hungary's automotive industry

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I. Importance and relevance

The automotive sector is one of the main pillars of Hungary's economy, contributing a major share to exports and employment, especially around Győr, Kecskemét and Debrecen. As the world moves toward electric vehicles (EVs) and software-heavy cars, domestic suppliers are under pressure to make more advanced, higher-value and knowledge-intensive components like batteries, electronics and sensors. The core question is whether the structure of firm-to-firm links - especially small-world features - helps companies learn faster and move up the economic complexity ladder, or whether it keeps them stuck in low-value activities.

II. Literature

Research on economic complexity shows that long-term growth depends on combining diverse capabilities to create more sophisticated products (Hidalgo & Hausmann, 2009; Hausmann & Hidalgo, 2011). In network theory, small-world structures help information and know-how spread quickly, though overly closed clusters can limit innovation (Fleming et al., 2007; Sorenson et al., 2006). These findings suggest that supplier network structures strongly influence whether firms can upgrade into more complex products. However, no similar study has yet examined this relationship in the context of Hungary, leaving an important research gap.

III. Research gap and question

Empirical evidence is still limited on which network characteristics, such as clustering or path length, actually support product upgrading at the firm level in Hungary. Therefore, this study asks: Does the small-world structure of Hungary's automotive supplier network increase the likelihood and speed of firms entering more complex, EV-related products? A possible approach is to measure network indicators at time $t - 1$ and estimate the probability that a firm moves into a product with a higher Product Complexity Index (PCI) at time t , controlling for firm size, age, region, and foreign ownership, etc.

IV. Stakeholders and potential benefits

The findings of this research could benefit several groups within Hungary's automotive ecosystem. Domestic SMEs could gain insights into how building connections across clusters and bridges can help them develop new capabilities and move toward higher-value production. For policy makers, the results could provide evidence to design more targeted supplier-development programs and better infrastructure that strengthen collaboration and innovation capacity. Finally, universities could use these insights to update their programs, placing greater emphasis on electronics, software, and quality engineering.

V. References

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